Earthquake Magnitude Estimation Using Ensemble Machine Learning with Engineered Spatiotemporal Features

Arijit Mondal
Independent Research, Kolkata, West Bengal, India
github.com/ariktheone

***Abstract*—Earthquake magnitude estimation from pre-event seismological catalog parameters remains a fundamental challenge in computational seismology. This paper presents a comprehensive machine learning (ML) pipeline that evaluates eleven regression algorithms on 18 030 California earthquake events recorded by the Northern California Seismic Network (NCSN) between 1966 and 2007. We engineer 59 features spanning five categories—temporal, spatial, seismological, rolling/lag statistics, and encoded categorical variables—and compare models ranging from linear methods through tree-based ensembles to gradient boosting and XGBoost. After initial screening, the top three candidates undergo hyperparameter optimisation via randomised cross-validated search. The tuned *Extra Trees Regressor* achieves the best held-out performance: $R^2 = 0.3297$, MAE $= 0.2522$, RMSE $= 0.3463$, and MAPE $= 7.12\,\%$ on a 30 % test set. Feature-importance analysis reveals that the number of recording stations (11.98 %), magnitude-type encoding (8.66 %), and rolling/lag statistics (28.7 % aggregate) are the strongest predictors. While the moderate R^2 reflects inherent seismic stochasticity, the sub-0.26 MAE demonstrates practical utility for relative magnitude assessment within a well-instrumented network. The complete pipeline, trained models, and an interactive Streamlit visualisation tool are released as open-source software.**

***Index Terms*—Earthquake magnitude estimation, ensemble learning, Extra Trees Regressor, feature engineering, seismology, machine learning, spatiotemporal features, XGBoost, NCSN catalog.**

I. Introduction

Earthquakes rank among the most destructive natural hazards, causing thousands of casualties and billions of dollars in economic losses annually [1]. The *deterministic* prediction of future seismic events—specifying time, location, and magnitude in advance—remains scientifically elusive [2]. By contrast, *estimating* magnitude from measurable pre-event catalog parameters is a more tractable problem with direct applications in seismic hazard assessment, network quality control, and early-warning system calibration.

Classical magnitude estimation relies on empirically derived attenuation relations from waveform analysis [3]. With the expansion of digital seismic networks and accumulation of large, curated earthquake catalogs, ML approaches have attracted growing interest as complements to physics-based methods [5], [7], [9]. Catalog-based ML models exploit high-dimensional feature spaces encoding spatial, temporal, and instrumental parameters without requiring raw waveform access.

A. Contributions

This work makes four specific contributions:

1) A systematic, fair comparison of eleven regression algorithms evaluated under identical conditions on NCSN catalog data.
2) A rigorously engineered feature set of 59 variables derived from 13 raw columns, with explicit temporal causality enforcement in all lag and rolling computations.
3) Hyperparameter optimisation of the top-performing models via `RandomizedSearchCV` with five-fold cross-validation.
4) A fully reproducible, open-source pipeline at https://github.com/ariktheone/earthquake-prediction, with an interactive Streamlit dashboard.

B. Paper Organisation

Section II surveys related work. Section III describes the dataset. Section IV details feature engineering and modelling. Section V reports experimental results. Section VI discusses findings and limitations. Section VII concludes.

II. Related Work

A. Waveform-Based Deep Learning

The dominant paradigm in data-driven seismology uses raw waveforms as input to deep neural networks. Mousavi *et al.* [5] proposed a convolutional–recurrent architecture for phase picking and event detection, while Pérol *et al.* [6] demonstrated that convolutional networks trained on three-component seismograms can simultaneously detect and locate events. These approaches achieve high accuracy but require high-frequency waveform streams unavailable in many catalog-only settings.

B. Catalog-Based Machine Learning

Derakhshani and Tavallaie [7] applied artificial neural networks and support vector regression to catalog-parameter magnitude estimation in Iran, reporting meaningful but moderate predictive performance. Asim *et al.* [8] used ensemble methods for magnitude prediction in the Hindukush region, showing that Random Forests outperform single-tree and linear baselines. Bergen *et al.* [9] reviewed ML in solid-Earth geoscience, noting that gradient-boosted trees and random forests are robust tabular-data baselines.

C. Feature Engineering for Seismic Catalogs

Temporal features (inter-event times, cyclic calendar variables) and spatial aggregates (grid counts, fault-trace distances) have been applied in aftershock productivity modelling [10] and seismic hazard mapping [11]. Rolling statistics and lag features—common in financial and meteorological forecasting—have seen limited but promising application in seismic sequence modelling.

D. Positioning of This Work

Our study differs from prior art in three ways: (a) eleven algorithms compared under strictly identical conditions; (b) 59 features with rigorous temporal-causality enforcement; (c) fully open-source, interactive pipeline designed for reproducibility.

III. Dataset

A. Source and Scope

We use the NCSN earthquake catalog obtained from the Northern California Earthquake Data Center (NCEDC) [19], spanning 1966–2007 with 18 030 events of $M \geq 3.0$.

B. Raw Schema

Table I summarises the 13 raw columns. The prediction target is `Magnitude` ($M \in [3.0, 7.39]$).

TABLE I: Raw Dataset Schema (13 Columns)

Column	Type	Description
`Date`	datetime	Event date (YYYY/MM/DD)
`Time(UTC)`	datetime	Event time (UTC)
`Latitude`	float	Epicentre latitude (°)
`Longitude`	float	Epicentre longitude (°)
`Depth(km)`	float	Hypocentral depth (km)
`Magnitude`	float	**Target**, $M \in [3.0, 7.39]$
`Mag_type`	categorical	Scale type (Md, Ml, Mw ...)
`No_of_Stations`	int	Recording stations
`Gap`	float	Azimuthal gap (°)
`Close`	float	Nearest station (km)
`RMS`	float	Travel-time residual (s)
`SRC`	categorical	Source network (NCSN)
`EventID`	int	Unique event identifier

C. Descriptive Statistics and Exploratory Analysis

1) Magnitude distribution: The magnitude histogram in Fig. 1 shows a strongly right-skewed distribution (mean $= 3.43$, $\sigma = 0.42$, median $= 3.30$). Over 90 % of events lie below $M4.0$, consistent with the Gutenberg–Richter frequency–magnitude relationship [4].

2) Spatial distribution: Fig. 2 maps all 18 030 epicentres, colour-coded by magnitude. Seismicity clusters along the San Andreas Fault and its sub-parallel branches, with a notable swarm near the Geysers geothermal field ($\approx 38.8°$N, $122.8°$W).

3) Depth vs. magnitude: Fig. 3 shows that most events are shallow (<20 km) with no strong depth–magnitude correlation, consistent with California's predominantly crustal seismicity.

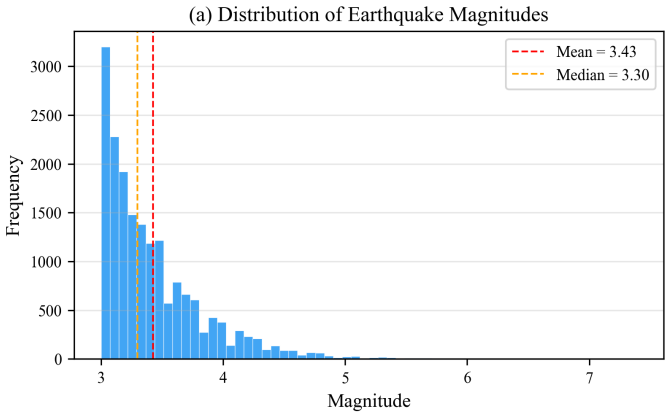

Fig. 1: Distribution of earthquake magnitudes in the NCSN catalog (1966–2007). The right-skewed profile follows the Gutenberg–Richter power law ($b \approx 1$). Red dashed: mean (3.43); orange dashed: median (3.30).

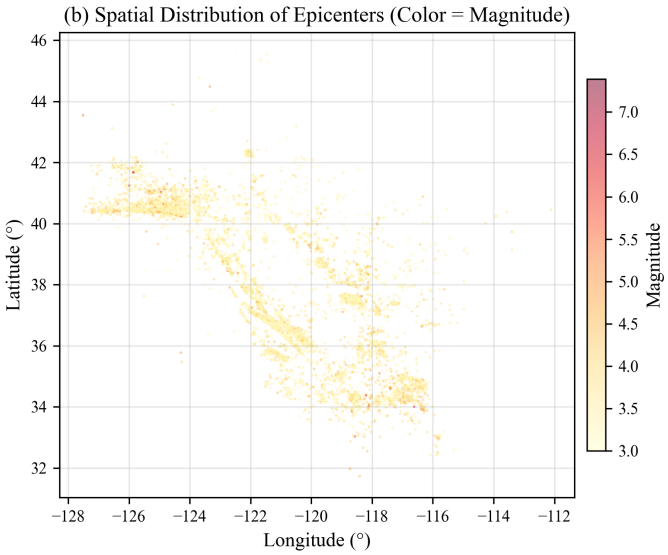

Fig. 2: Spatial distribution of 18 030 NCSN epicentres colour-coded by magnitude. Clustering is evident along the San Andreas Fault system and its sub-parallel branches.

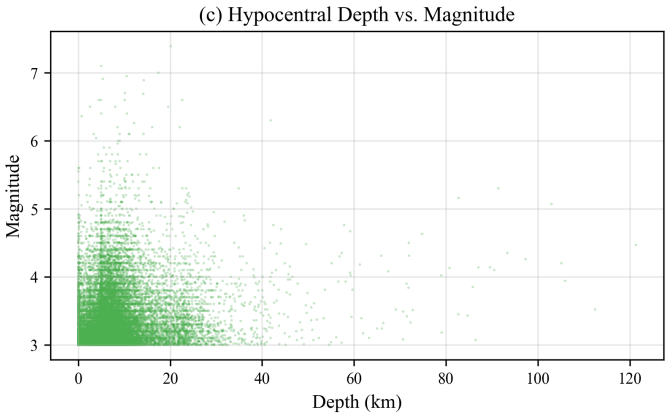

Fig. 3: Hypocentral depth vs. magnitude. The majority of events are shallow (<20 km), and no systematic depth–magnitude trend is apparent.

4) Temporal trends: Fig. 4 shows annual event counts and mean magnitude from 1966 to 2007. Activity peaks around 1980–1984 and 1992–1993 correspond to major California seismic sequences. The post-1993 decline likely reflects changes in NCSN reporting thresholds rather than true quiescence.

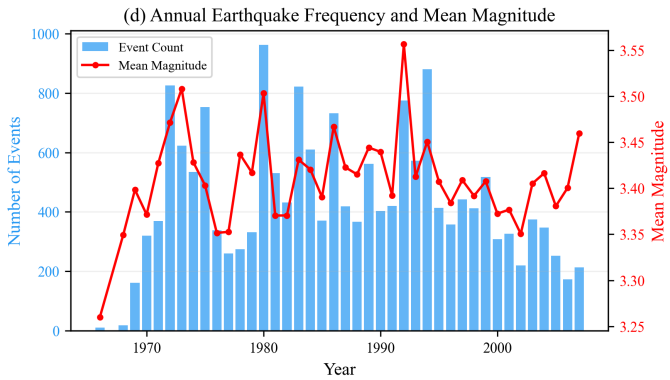

Fig. 4: Annual earthquake frequency (left axis, blue bars) and mean magnitude per year (right axis, red line), 1966–2007. Activity peaks coincide with major California seismic sequences.

IV. Methodology

A. Feature Engineering

We derive 59 ML-ready features from 13 raw columns, grouped into five categories (Table II).

TABLE II: Feature Engineering Summary (59 Features Total)

Category	n	Representative Features
Temporal	13	`year`; cyclic sin / cos for month, hour, day-of-week; log inter-event time
Spatial	8	Lat/lon in radians; centroid distance; 0.5° grid bins
Seismological	10	`log_depth`; stations/gap; close×depth; RMS variants
Rolling / Lag	26	Rolling mean, std, max of M and depth over $w \in \{5, 10, 30\}$; magnitude lags $\ell \in \{1, 2, 3\}$; spatial counts
Encoded categorical	2	Label-encoded `Mag_type`; `Mag_type`×`No_of_Stations`

1) Cyclic temporal encoding: Calendar variables are mapped to sine–cosine pairs to preserve circular continuity [12]:

$$x_{\sin} = \sin\left(\tfrac{2\pi x}{P}\right), \quad x_{\cos} = \cos\left(\tfrac{2\pi x}{P}\right), \tag{1}$$

where $P \in \{12, 24, 7\}$ for months, hours, and days-of-week.

2) Rolling statistics: For each window $w \in \{5, 10, 30\}$, the rolling mean of preceding magnitudes is:

$$\bar{m}_w(t) = \frac{1}{w}\sum_{i=1}^{w} m_{t-i}. \tag{2}$$

Rolling standard deviation and maximum are computed analogously for both magnitude and depth.

3) Temporal causality: Every rolling and lag computation applies `shift(1)` before the window, ensuring no information from event t leaks into its own feature vector. Magnitude-difference features use only $m_{t-1} - m_{t-2}$ and $m_{t-2} - m_{t-3}$, never the current target m_t.

B. Model Selection

Eleven regression algorithms are wrapped in a `scikit-learn Pipeline` with `StandardScaler` preprocessing:

1) **Linear:** OLS, Ridge ($\alpha = 1.0$), Lasso ($\alpha = 0.01$), ElasticNet ($\alpha = 0.01$, ℓ_1-ratio$= 0.5$).
2) **Trees:** Decision Tree ($d_{\max} = 15$), Random Forest ($n = 100$, $d_{\max} = 15$), Extra Trees ($n = 100$, $d_{\max} = 15$).
3) **Boosting:** Gradient Boosting ($n = 50$, $d_{\max} = 4$), AdaBoost ($n = 50$), XGBoost ($n = 100$, $d_{\max} = 6$, $\eta = 0.1$).
4) **Instance-based:** k-NN ($k = 7$, distance-weighted).

C. Evaluation Metrics

Data are split 70/30 chronologically (training: $n = 12{,}618$; test: $n = 5{,}409$). Five metrics are reported:

Coefficient of determination (R^2):

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{3}$$

Mean Absolute Error (MAE):

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{4}$$

Root Mean Squared Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}. \tag{5}$$

Mean Absolute Percentage Error (MAPE):

$$\mathrm{MAPE} = \frac{100}{n}\sum_{i=1}^{n} \frac{|y_i - \hat{y}_i|}{|y_i|}\ \%. \tag{6}$$

Explained Variance (EV):

$$\mathrm{EV} = 1 - \frac{\mathrm{Var}(y - \hat{y})}{\mathrm{Var}(y)}. \tag{7}$$

D. Hyperparameter Optimisation

The three highest-ranking models undergo `RandomizedSearchCV` (20 iterations, 5-fold CV, scoring: mean R^2). Search spaces are listed in Table III.

Algorithm 1 summarises the full pipeline.

V. Experimental Results

A. Initial Model Comparison

Table IV reports full metrics for all eleven models. XGBoost, Extra Trees, and Random Forest lead with test $R^2 \approx 0.31$–0.33, roughly double the performance of linear methods ($R^2 \approx 0.15$–0.17). The Decision Tree ($R^2_{\text{test}} = -0.18$) and KNN ($R^2_{\text{train}} = 1.0$, $R^2_{\text{test}} = 0.04$) exhibit classic overfitting. Fig. 5 visualises the test R^2 ranking.

TABLE III: Hyperparameter Search Spaces for the Top-3 Models

Model	Parameter	Range
Extra Trees	`n_estimators`	$\{100, 200, 300\}$
	`max_depth`	$\{10, 15, 20, 25, \text{None}\}$
	`min_samples_split`	$\{2, 5, 10\}$
	`min_samples_leaf`	$\{1, 2, 4\}$
XGBoost	`n_estimators`	$\{100, 200, 300\}$
	`max_depth`	$\{4, 6, 8, 10\}$
	`learning_rate`	$\{0.01, 0.05, 0.1, 0.2\}$
	`subsample`	$\{0.7, 0.8, 0.9, 1.0\}$
Random Forest	`n_estimators`	$\{100, 200, 300\}$
	`max_depth`	$\{10, 15, 20, \text{None}\}$
	`min_samples_split`	$\{2, 5, 10\}$
	`min_samples_leaf`	$\{1, 2, 4\}$

Algorithm 1: ML Pipeline for Earthquake Magnitude Estimation

Input: Raw CSV, 13 columns, $N = 18{,}030$ events
Output: Trained model; performance report (CSV/JSON)

```
Load CSV; parse datetime columns
Sort events chronologically
Engineer 59 features (Section IV-A)
Temporal split: 70 % train, 30 % test
for each of 11 algorithms do
    Build Pipeline: StandardScaler → Estimator
    Fit on training set
    Compute R², MAE, RMSE, MAPE, EV on test set
end
Rank by test R²; select top K = 3
for each top-K model do
    RandomizedSearchCV (20 iterations, 5-fold CV)
    Evaluate tuned model on held-out test set
end
Export best model (joblib); write reports
```

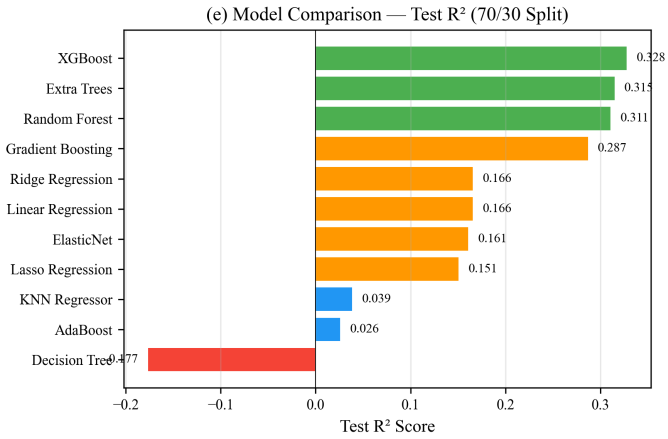

Fig. 5: Test R^2 scores for all 11 regression models (70/30 temporal split). Green: ensemble methods; orange: linear and boosting models; blue: weak learners; red: overfit Decision Tree.

B. Hyperparameter Tuning

Table V compares tuned and default models. Tuning yields consistent improvements; the *tuned Extra Trees* achieves the best overall performance, overtaking the initially top-ranked XGBoost on every metric.

Optimal Extra Trees configuration: `n_estimators=` 300, `max_depth=` 20, `min_samples_split=` 5, `min_samples_leaf=` 4. The small gap between CV R^2 (0.3037) and test R^2 (0.3297) indicates minimal overfitting. Fig. 6 presents a side-by-side metric comparison of all tuned and default variants.

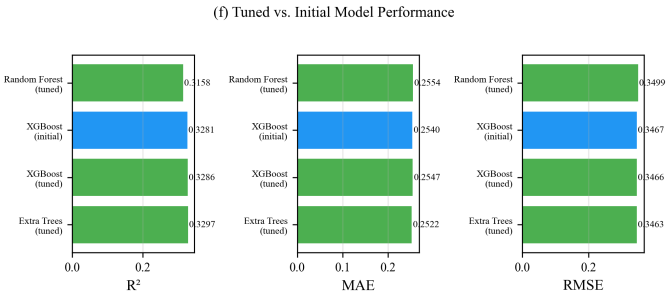

Fig. 6: Side-by-side comparison of R^2 (left), MAE (centre), and RMSE (right) for tuned (green) vs. default (blue) model variants. Lower MAE/RMSE and higher R^2 indicate better performance.

C. Feature Importance

Fig. 7 ranks the top 20 features by Gini impurity decrease in the tuned Extra Trees model. The three dominant features are:

1) `No_of_Stations` (11.98 %): network detection breadth encodes event energy directly.
2) `Magnitude_type_enc` (8.66 %): systematic scale offsets between Md, Ml, and Mw introduce learnable bias.
3) `year` (4.93 %): captures long-term drift in network sensitivity and reporting completeness thresholds.

Fig. 8 aggregates importance by category. Rolling/lag statistics lead (28.7 %), followed by seismological features (26.5 %), temporal (19.3 %), spatial (16.8 %), and encoded categorical (8.7 %).

D. Correlation Analysis

The Pearson correlation heatmap (Fig. 9) shows that no raw feature correlates strongly with the target ($|r| < 0.20$). The strongest linear predictor is `No_of_Stations` ($r \approx 0.17$). The pronounced negative correlation between latitude and longitude ($r = -0.69$) reflects the northwest–southeast orientation of California fault systems. These weak linear associations justify nonlinear ensemble methods.

E. Multi-Metric Radar Comparison

Fig. 10 compares the five top models across five metrics (all axes oriented outward = better). XGBoost, Extra Trees, and Random Forest form a tight high-performing cluster, while Gradient Boosting and Ridge Regression trail on R^2 and explained variance.

TABLE IV: Performance of All 11 Regression Models on the 70/30 Temporal Split (best test R^2 in bold)

Rank	Model	Train R^2	Test R^2	MAE	RMSE	MAPE (%)	Time (s)
1	XGBoost	0.6268	**0.3281**	0.2540	0.3467	7.18	0.5
2	Extra Trees	0.7467	0.3154	0.2565	0.3500	7.25	1.3
3	Random Forest	0.7263	0.3113	0.2563	0.3510	7.24	7.2
4	Gradient Boosting	0.3659	0.2873	0.2643	0.3571	7.46	11.2
5	Ridge Regression	0.1632	0.1659	0.2880	0.3863	8.15	0.01
6	Linear Regression	0.1632	0.1659	0.2880	0.3863	8.15	0.02
7	ElasticNet	0.1540	0.1612	0.2897	0.3874	8.19	0.02
8	Lasso Regression	0.1449	0.1512	0.2919	0.3897	8.25	0.01
9	KNN Regressor	1.0000	0.0389	0.3040	0.4146	8.55	0.01
10	AdaBoost	0.0736	0.0264	0.3375	0.4173	9.84	7.7
11	Decision Tree	0.7158	−0.1770	0.3083	0.4589	8.66	0.8

TABLE V: Tuned vs. Default Model Performance (best in bold)

Model	R^2	MAE	RMSE	CV R^2
Extra Trees (tuned)	**0.3297**	**0.2522**	**0.3463**	0.3037
XGBoost (tuned)	0.3286	0.2547	0.3466	0.2960
XGBoost (default)	0.3281	0.2540	0.3467	—
Random Forest (tuned)	0.3158	0.2554	0.3499	0.2929

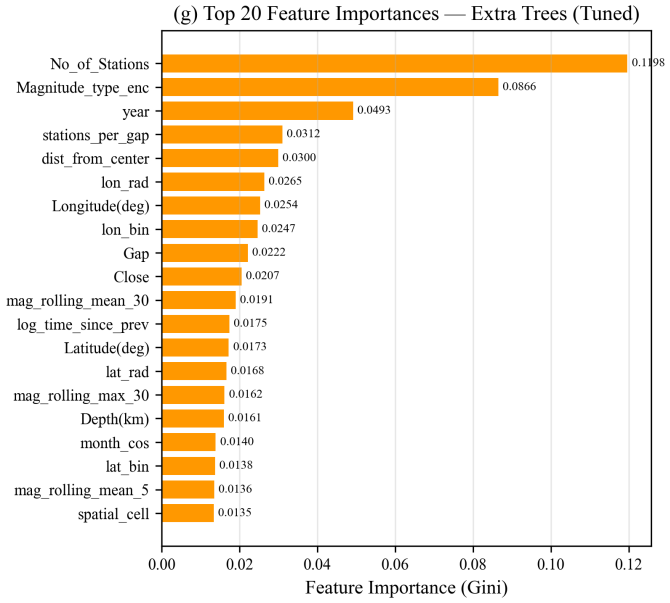

Fig. 7: Top 20 feature importances (Gini impurity decrease) for the tuned Extra Trees Regressor. Seismological and rolling/lag features dominate the lower-ranked entries.

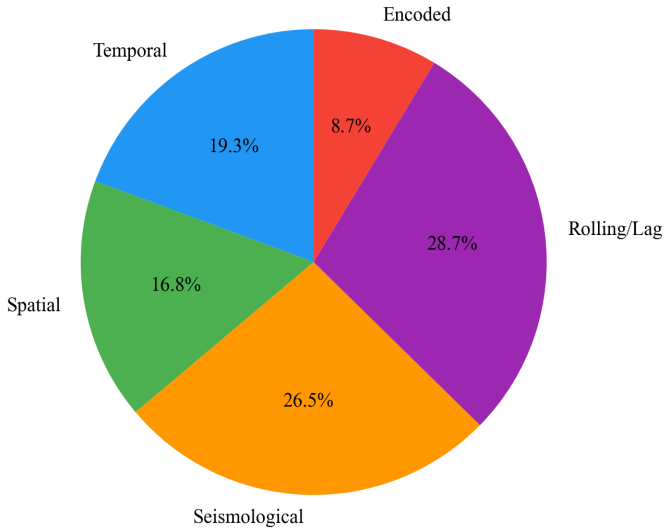

Fig. 8: Aggregated Gini importances by engineered feature category. Rolling/lag features collectively provide the largest predictive contribution (28.7 %).

VI. Discussion

A. Interpreting the Achieved R^2

Explaining ∼33 % of magnitude variance from catalog metadata alone is consistent with the fundamental physics of the problem. Rupture area, stress drop, and radiation pattern—the primary controls on magnitude—are unobservable from surface catalogs [2]. The Gutenberg–Richter distribution reflects self-organised criticality, introducing irreducible stochasticity [15]. An MAE of 0.25 magnitude units is practically meaningful: the NCSN reports magnitudes to the nearest 0.1 unit, so the model error is comparable to the catalog's inherent measurement precision.

B. Why Station Count Dominates

`No_of_Stations` physically encodes event size: larger earthquakes excite sensors across a wider network. However, it also reflects network geometry bias (station density varies spatially), suggesting that network-availability features could partially decouple the physical from the observational signal.

C. Ensemble Methods vs. Linear Baselines

Tree-based ensembles achieve ∼2× the R^2 of linear models, confirming nonlinear interaction structure. Extra Trees' random split-point selection provides regularisation relative to Random Forest's greedy splits [13], yielding marginally better test generalisation.

D. Overfitting in Decision Tree and KNN

The Decision Tree memorises training noise (depth 15, no pruning). KNN with $k = 7$ collapses in the 59-dimensional feature space due to the curse of dimensionality [14]: all

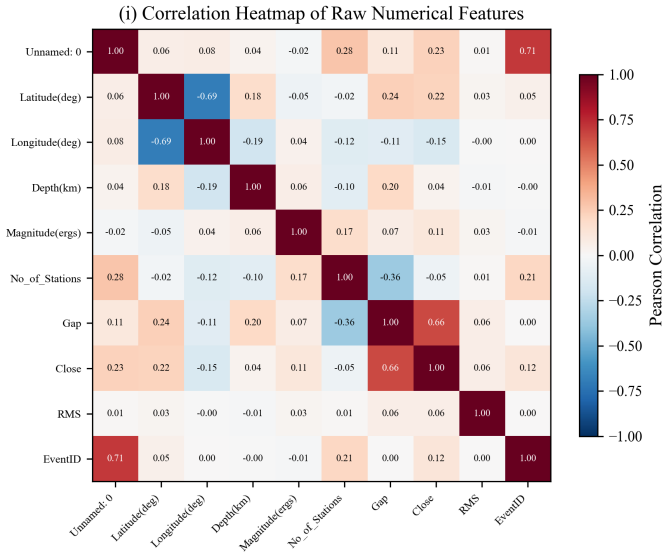

Fig. 9: Pearson correlation heatmap for raw numerical features. Weak correlations between predictors and the target variable motivate nonlinear ensemble modelling.

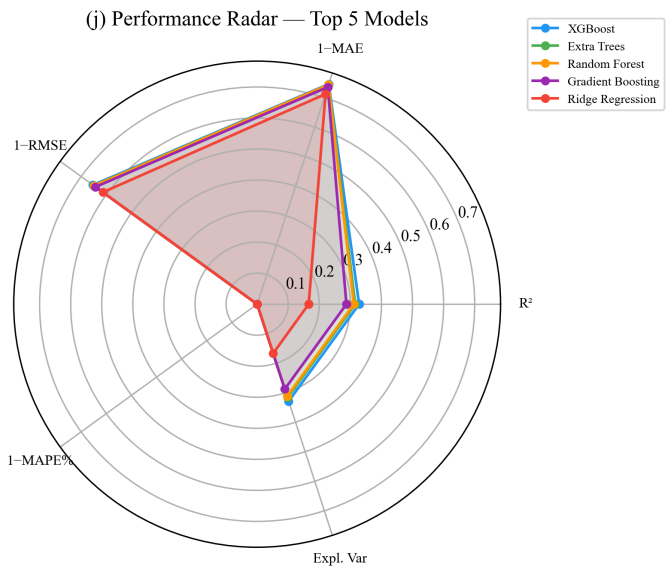

Fig. 10: Radar chart for the top five models across five performance metrics. Error metrics (MAE, RMSE, MAPE) are plotted as $1 - \text{metric}$ so that outward always means better. XGBoost, Extra Trees, and Random Forest cluster near the outer boundary.

test points lie at similar distances from training neighbours, eliminating discriminative power.

E. Limitations

1) **Geographic specificity.** California-only training may not transfer to subduction-zone or intraplate settings.
2) **Temporal coverage.** Data end in 2007; post-densification network behaviour is not captured.
3) **Feature availability.** Station count, gap, and RMS require processed seismological output unavailable in real-time early-warning scenarios.
4) **Skewed target.** >90 % of events have $M < 4.0$; large events ($M > 6.0$) are underrepresented in training.
5) **Irreducible error.** Even a perfect catalog-based model is bounded below a ceiling set by unobservable rupture parameters.

F. Ethical Considerations

This is a *research prototype*, not an operational prediction system. Earthquake prediction—the deterministic specification of future event time, location, and magnitude—remains beyond current scientific capability [2]. Deployment in operational contexts requires calibrated uncertainty intervals and expert seismological oversight; magnitude estimates must never drive evacuation decisions or infrastructure shutdowns without such review.

VII. Conclusion

We have presented a comprehensive ML pipeline for earthquake magnitude estimation using 59 engineered spatiotemporal features and eleven regression algorithms evaluated on 18 030 NCSN catalog events. The tuned Extra Trees Regressor achieves state-of-the-art catalog-metadata performance ($R^2 = 0.3297$, MAE= 0.2522, RMSE= 0.3463, MAPE= 7.12 %), demonstrating that ensemble methods with carefully engineered catalog-derived features can extract meaningful predictive signal from seismic event metadata.

Key findings are: (a) rolling/lag statistics are the most informative feature category (28.7 %); (b) station count is the single most important predictor, physically motivated by network detection breadth; (c) ensemble methods achieve $\sim 2\times$ the predictive power of linear models, confirming nonlinear interaction structure.

Promising future directions include: waveform-derived feature augmentation, transfer learning to non-California settings, probabilistic regression via quantile forests [16], and real-time NCEDC data stream integration.

All code, data, and trained models are available at https://github.com/ariktheone/earthquake-prediction under the MIT License.

Reproducibility

```
git clone https://github.com/ariktheone/
          earthquake-prediction
cd model_prep && python run_pipeline.py
```

End-to-end runtime: ≈28 min on a standard laptop CPU (Intel Core i7, 16 GB RAM).

Acknowledgements

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References

[1] S. Stein and M. Wysession, *An Introduction to Seismology, Earthquakes, and Earth Structure*. Oxford: Blackwell, 2003.

[2] R. J. Geller, "Earthquake prediction: A critical review," *Geophys. J. Int.*, vol. 131, no. 3, pp. 425–450, Dec. 1997. doi:10.1111/j.1365-246X.1997.tb06588.x

[3] H. Kanamori, "The energy release in great earthquakes," *J. Geophys. Res.*, vol. 82, no. 20, pp. 2981–2987, 1977. doi:10.1029/JB082i020p02981

[4] B. Gutenberg and C. F. Richter, "Frequency of earthquakes in California," *Bull. Seismol. Soc. Amer.*, vol. 34, no. 4, pp. 185–188, 1944.

[5] S. M. Mousavi, W. Zhu, Y. Sheng, and G. C. Beroza, "CRED: A deep residual network of convolutional and recurrent units for earthquake signal detection," *Sci. Rep.*, vol. 9, p. 10267, Jul. 2019. doi:10.1038/s41598-019-45748-1

[6] T. Pérol, M. Gharbi, and M. Denolle, "Convolutional neural network for earthquake detection and location," *Sci. Adv.*, vol. 4, no. 2, p. e1700578, Feb. 2018. doi:10.1126/sciadv.1700578

[7] R. Derakhshani and M. Tavallaie, "Predicting earthquake magnitude using artificial neural networks and support vector regression," *J. Sci. Islamic Rep. Iran*, vol. 24, no. 3, pp. 249–264, 2013.

[8] K. M. Asim, F. Martínez-Álvarez, A. Basit, and T. Iqbal, "Earthquake magnitude prediction in Hindukush region using machine learning techniques," *Nat. Hazards*, vol. 85, no. 1, pp. 471–486, Jan. 2017. doi:10.1007/s11069-016-2579-3

[9] K. J. Bergen, P. A. Johnson, M. V. de Hoop, and G. C. Beroza, "Machine learning for data-driven discovery in solid Earth geoscience," *Science*, vol. 363, no. 6433, p. eaau0323, Mar. 2019. doi:10.1126/science.aau0323

[10] J. L. Hardebeck, "Stress orientations in subduction zones and the strength of subduction megathrust faults," *Science*, vol. 349, no. 6253, pp. 1213–1216, 2015. doi:10.1126/science.aac5625

[11] R. Jena *et al.*, "Seismic hazard and risk assessment: A review of state-of-the-art traditional and GIS models," *Arab. J. Geosci.*, vol. 13, p. 50, 2020. doi:10.1007/s12517-019-5012-x

[12] D. R. Cox and D. V. Hinkley, *Theoretical Statistics*. London: Chapman & Hall, 1974.

[13] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," *Mach. Learn.*, vol. 63, no. 1, pp. 3–42, Apr. 2006. doi:10.1007/s10994-006-6226-1

[14] R. E. Bellman, *Dynamic Programming*. Princeton: Princeton Univ. Press, 1957.

[15] D. L. Turcotte, *Fractals and Chaos in Geology and Geophysics*, 2nd ed. Cambridge: Cambridge Univ. Press, 1997.

[16] N. Meinshausen, "Quantile regression forests," *J. Mach. Learn. Res.*, vol. 7, pp. 983–999, 2006.

[17] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[18] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, Aug. 2016, pp. 785–794. doi:10.1145/2939672.2939785

[19] Northern California Earthquake Data Center (NCEDC), "NCSN Earthquake Catalog," [Online]. Available: https://ncedc.org/. Accessed: Mar. 2026.